Project Initialization and Planning Phase

Date	05 October 2026	
Team ID	SWTID-2026-1369	
Project Title	AI Blog Nest API	Team Leader: Priyadharshini S Team Members: Monica L, Durga K, Mothieswari R
Maximum Marks		

Project Proposal (Proposed Solution) report

AI Blog Nest API is a developer-friendly API for creating structured blog content with AI. It reduces manual drafting effort and provides a consistent content workflow.

Project Overview

Objective: Provide a simple API that accepts a topic and returns an organized blog draft with title, sections, summary, and tags.

Scope: API request handling, AI content generation, structured JSON responses, validation, error handling, and optional storage of generated posts.

Problem Statement

Description: Developers and content teams often need repeated blog creation, but manual drafting is slow and inconsistent across formats.

Impact: A unified API can reduce drafting time, standardize output structure, and simplify integration into websites or applications.

Proposed Solution

Approach: Build a REST API with authentication, request validation, AI generation, structured responses, and documentation.

Key Features: topic-to-blog generation; title and summary generation; tags; JSON API response; validation; error handling; API documentation.

	AI generation endpoint; structured JSON response; validation; secure API access; logging and error handling.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifica number of cores	CPU suitable for API server; optional GPU/hosted AI model
Memory	RAM specifications	8 GB RAM recommended for development
Storage	Disk space for data, and logs	20 GB+ SSD for app, logs and local data
Software		
Frameworks	Python frameworks	FastAPI / Python
Libraries	Additional libraries	Pydantic, httpx, SQLAlchemy, AI SDK
Development Environment	IDE	VS Code / PyCharm
Data		
Data	Source, size, format	Test blog dataset / sample JSON requests